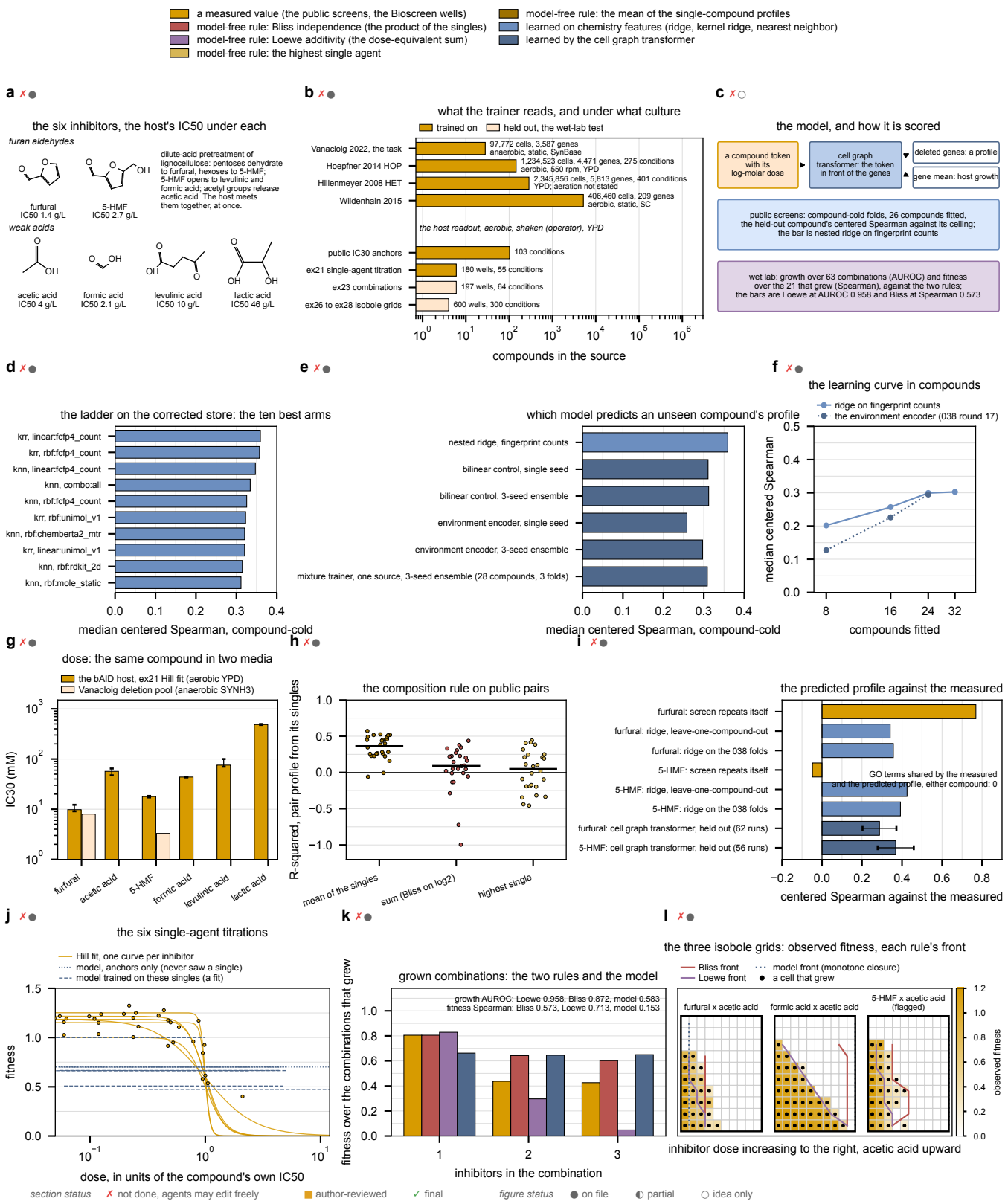

a measured value (the public screens, the Bioscreen wells)
model-free rule: Bliss independence (the product of the singles)
model-free rule: Loewe additivity (the dose-equivalent sum)
model-free rule: the highest single agent
model-free rule: the mean of the single-compound profiles
learned on chemistry features (ridge, kernel ridge, nearest neighbor)
learned by the cell graph transformer
a
the six inhibitors, the host's IC50 under each
furan aldehydes
furfural IC50 1.4 g/L
5-HMF IC50 2.7 g/L
dilute-acid pretreatment of lignocellulose: pentoses dehydrate to furfural, hexoses to 5-HMF; 5-HMF opens to levulinic and formic acid; acetyl groups release acetic acid. The host meets them together, at once.
weak acids
acetic acid IC50 4 g/L
formic acid IC50 2.1 g/L
levulinic acid IC50 10 g/L
lactic acid IC50 46 g/L
b
what the trainer reads, and under what culture
trained on
held out, the wet-lab test
Vanacloig 2022, the task
97,772 cells, 3,587 genes
anaerobic, static, SynBase
Hoepfner 2014 HOP
1,234,523 cells, 4,471 genes, 275 conditions
aerobic, 550 rpm, YPD
Hillenmeyer 2008 HET
2,345,856 cells, 5,813 genes, 401 conditions
YPD; aeration not stated
Wildenhain 2015
406,460 cells, 209 genes
aerobic, static, SC
the host readout, aerobic, shaken (operator), YPD
public IC30 anchors
103 conditions
ex21 single-agent titration
180 wells, 55 conditions
ex23 combinations
197 wells, 64 conditions
ex26 to ex28 isobole grids
600 wells, 300 conditions
compounds in the source
c
the model, and how it is scored
a compound token with its log-molar dose
cell graph transformer: the token in front of the genes
deleted genes: a profile
gene mean: host growth
public screens: compound-cold folds, 26 compounds fitted, the held-out compound's centered Spearman against its ceiling; the bar is nested ridge on fingerprint counts
wet lab: growth over 63 combinations (AUROC) and fitness over the 21 that grew (Spearman), against the two rules; the bars are Loewe at AUROC 0.958 and Bliss at Spearman 0.573
d
the ladder on the corrected store: the ten best arms
krr, linear:fcfp4_count
krr, rbf:fcfp4_count
knn, linear:fcfp4_count
knn, combo:all
knn, rbf:fcfp4_count
krr, rbf:unimol_v1
knn, rbf:chemberta2_mtr
krr, linear:unimol_v1
knn, rbf:rdkit_2d
knn, rbf:mole_static
median centered Spearman, compound-cold
e
which model predicts an unseen compound's profile
nested ridge, fingerprint counts
bilinear control, single seed
bilinear control, 3-seed ensemble
environment encoder, single seed
environment encoder, 3-seed ensemble
mixture trainer, one source, 3-seed ensemble (28 compounds, 3 folds)
median centered Spearman, compound-cold
f
the learning curve in compounds
ridge on fingerprint counts
the environment encoder (038 round 17)
median centered Spearman
compounds fitted
g
dose: the same compound in two media
the bAID host, ex21 Hill fit (aerobic YPD)
Vanacloig deletion pool (anaerobic SYNH3)
IC30 (mM)
furfural
acetic acid
5-HMF
formic acid
levulinic acid
lactic acid
h
the composition rule on public pairs
R-squared, pair profile from its singles
mean of the singles
sum (Bliss on log2)
highest single
i
the predicted profile against the measured
furfural: screen repeats itself
furfural: ridge, leave-one-compound-out
furfural: ridge on the 038 folds
5-HMF: screen repeats itself
5-HMF: ridge, leave-one-compound-out
5-HMF: ridge on the 038 folds
furfural: cell graph transformer, held out (62 runs)
5-HMF: cell graph transformer, held out (56 runs)
GO terms shared by the measured and the predicted profile, either compound: 0
centered Spearman against the measured
j
the six single-agent titrations
Hill fit, one curve per inhibitor
model, anchors only (never saw a single)
model trained on these singles (a fit)
fitness
dose, in units of the compound's own IC50
k
grown combinations: the two rules and the model
growth AUROC: Loewe 0.958, Bliss 0.872, model 0.583
fitness Spearman: Bliss 0.573, Loewe 0.713, model 0.153
fitness over the combinations that grew
inhibitors in the combination
l
the three isobole grids: observed fitness, each rule's front
Bliss front
Loewe front
model front (monotone closure)
a cell that grew
furfural x acetic acid
formic acid x acetic acid
5-HMF x acetic acid (flagged)
observed fitness
inhibitor dose increasing to the right, acetic acid upward
section status
not done, agents may edit freely
author-reviewed
final
figure status
on file
partial
idea only